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EXPLAINING INTERVIEWER EFFECTS: A RESEARCH SYNTHESIS

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A rich and diverse literature exists on the effects that human interviewers can have on different aspects of the survey data collection process. This research synthesis uses the Total Survey Error (TSE) framework to highlight important historical developments and advances in the study of interviewer effects on a variety of important survey process outcomes, including sample frame coverage, contact and recruitment of potential respondents, survey measurement, and data processing. Included in the scope of the synthesis is research literature that has focused on *explaining* variability among interviewers in these effects and the different types of variable errors that they can introduce, which can ultimately affect the efficiency of survey estimates. We first consider common tasks with which human interviewers are often charged and then use the TSE framework to organize and synthesize the literature discussing the variable errors that interviewers can introduce when attempting to execute each task. Based on our synthesis, we identify key gaps in knowledge and then use these gaps to motivate an organizing model for future research investigating explanations for interviewer effects on different aspects of the survey data collection process.

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1. INTRODUCTION

Although interviewing practices were certainly different in the early-to-mid twentieth century compared with today (Kahn and Cannell 1957; Converse 2011), historical perspectives on interviewer effects have informed our current understanding. We now know that interviewers affect the survey process both positively (e.g., increasing response rates) and negatively (e.g., introducing measurement errors). The Total Survey Error (TSE) framework (Groves, Fowler, Couper, Lepkowski, Singer, et al. 2009) describes the different ways that interviewers can affect the survey process. With this research synthesis, we attempt to synthesize the findings of studies that have attempted to *explain the variability* among interviewers in coverage, nonresponse, measurement, and processing errors using interviewer-level characteristics.

Interviewer variability in measurement errors, for instance, introduces non-zero correlations among the responses collected by an interviewer. These within-interviewer correlations inflate the variance of descriptive survey estimates by a factor of approximately $1 + \rho_{\text{int}}(m - 1)$, where m is the average number of interviews conducted by an interviewer and ρ_{int} is the within-interviewer correlation (Groves 2004). An average interviewer workload of 35 respondents and a within-interviewer correlation of 0.03 would thus double the estimated variance of a mean or effectively halve the sample size (where $n_{eff} = n/(1 + \rho_{\text{int}}(m - 1))$) . This underscores the importance of identifying and understanding those interviewer-level characteristics that have strong relationships with different survey outcomes. Given this focus, we use the TSE framework to update and extend previous syntheses of the interviewer effects literature (e.g., Japec 2008; Schaeffer, Dykema, and Maynard 2010) as follows:

(1) We link recent studies of interviewer effects on measurement and nonresponse errors to previous work, synthesizing the results of *related studies* examining the associations between specific interviewer characteristics and these errors.
(2) We synthesize results from studies of interviewer effects on coverage errors and paradata, including indicators of consent to additional data collection.
(3) We synthesize results from studies of interviewer effects on processing error.
(4) We present an organizing model for guiding future studies of interviewer effects on multiple sources of survey error.

We specifically focus on literature that has attempted to *explain* interviewer variability in these effects or understand sources of the variability among interviewers in particular indicators of survey errors (e.g., response accuracy). While we also consider studies that have attempted to identify correlates of the *magnitude* of interviewer variance (e.g., Freeman and Butler 1976), we focus primarily on studies examining the relationships of interviewer-level characteristics with survey outcomes. We seek answers to the following questions:

(1) What are the known explanations for why interviewers vary in terms of key indicators of coverage, nonresponse, measurement, and processing errors?
(2) When considering *related studies* of the relationship between a specific interviewer characteristic and a particular indicator of survey error, what is the distribution of the results of these studies, and are underlying conceptual explanations apparent?
(3) What are the critical gaps in knowledge that need to be the focus of future research studies attempting to explain interviewer effects?

1.1 Approaches to Studying Interviewer Effects

Studies attempting to isolate interviewer effects must assume that interviewers have equivalent assignments in terms of the expected values of a variable of interest (Mahalanobis 1946; Hansen, Hurwitz, and Bershad 1960; Fellegi 1964; U.S. Bureau of the Census 1985), and there are different approaches used to meet this assumption in practice. *Fully interpenetrated designs* randomly assign sampled units to interviewers. Cost constraints and interviewers' working schedules seldom allow for such designs. In telephone surveys, *quasi-interpenetrated designs*, where an allocation algorithm randomly assigns sampled units to those interviewers currently working, mimic full interpenetration. In face-to-face surveys, which generally produce larger interviewer effects (Groves and Kahn 1979), interviewers are usually assigned to work in a single geographic area, confounding interviewer and area effects. To disentangle area effects and interviewer effects, researchers have implemented *partially interpenetrated designs*, where adjacent geographic areas are pooled and at least two interviewers are randomly assigned to sample units in each of those areas (e.g., O'Muircheartaigh and Campanelli 1998).

Studies using interpenetrated designs then account for *random* interviewer effects (typically, random intercepts) in multilevel models (Hox, de Leeuw, and Kreft 1991; Hox 1994), assuming that interviewers are randomly selected from larger hypothetical populations of interviewers. A failure to account for within-interviewer clustering in this fashion was a significant problem in earlier studies (Dijkstra 1983). In partially interpenetrated designs, the models generally include *crossed* random effects of interviewers and areas. Interviewer effects are then typically expressed through the intraclass correlation coefficient (ICC), or the proportion of the total variance in the outcome being modeled that is due to variance in the interviewer effects. If interpenetrated designs are not feasible (which is often the case in face-to-face surveys) or there is evidence of variability among interviewers in respondent characteristics despite random assignment (e.g., West and Olson 2010), researchers need to carefully specify multilevel models that include the fixed effects of both area and respondent characteristics (Schaeffer et al. 2010).

We aim to synthesize evidence from studies of both face-to-face and telephone surveys that have used these approaches *and* sought to explain interviewer variance by using *fixed* effects of measurable interviewer characteristics (e.g., experience). Many of these studies are *observational* in nature, where causal inferences related to explanatory factors are not possible in the absence of random assignment of these characteristics to interviewers.

1.2 Methodology

Table 1 presents the various tasks for which interviewers may be responsible and the corresponding types of variable errors that interviewers might introduce in the context of the TSE framework.

We organized the literature based on the order of the tasks and the corresponding errors in table 1. We used the following methodology:

(1) First, we used online search tools (e.g., Google Scholar) to identify published articles, technical reports, papers from conference proceedings, or book chapters attempting to explain variance in interviewer effects, using the search terms "interviewer," "effects," "variance," and "surveys" simultaneously.
(2) Second, we determined whether the identified research products (and those studies cited therein) sought to *explain* interviewer effects; if they did, we

Table 1 Common Interviewer Tasks and Survey Errors Potentially Introduced by Each Interviewer

Interviewer task	Survey errors potentially introduced
Generate sampling frames	Coverage error
Make contact	Unit nonresponse error
Gain cooperation	Unit nonresponse error
Gain consent to additional parts of the survey	Unit nonresponse error
Ask survey questions, conduct measurements, and maintain motivation	Measurement error, Item Nonresponse error
Record answers and measurements	Processing error

recorded the dependent variable(s) under investigation, the key interviewer-level predictor variables, and the ways in which these variables were measured. Studies reporting evidence of interviewer variance in particular outcomes but not attempting to explain this variance were excluded.

(3) Third, we systematically organized studies by combinations of dependent variable and explanatory factor and recorded measures of effect sizes, standard errors, confidence intervals, and *p* values (enabling syntheses of the findings of related studies).

While reviewing each area of the literature, we identified key gaps in knowledge and synthesized the findings related to important explanatory factors. Finally, we used our syntheses to develop an organizing model for guiding future research.

2. EXPLAINING INTERVIEWER EFFECTS ON COVERAGE ERROR

In face-to-face surveys, interviewers may help to develop frames for area probability samples by building or refining address lists. Kwiat (2009) and Eckman (2013) reported evidence of variability among interviewers in this process (e.g., different methods of listing addresses in multi-unit buildings). Eckman and Kreuter (2011) reported that many interviewers simply confirm what is on an existing address list rather than adding or deleting addresses. Eckman and O'Muircheartaigh (2011) suggested that some interviewers fail to add missing housing units when applying the half-open interval method, possibly to avoid increased workloads given response rate targets, and only add missing units when nearby units are eligible and/or cooperative. While improved standardized training on this listing process may help to reduce this variability in coverage *rates* among interviewers, whether this leads to variance among interviewers in actual coverage *errors* for key statistics (Groves 2004) remains unknown.

Interviewers also conduct screening interviews and/or generate household rosters, specifically for eventual (computer-automated) within-household sampling of individuals. Tourangeau, Shapiro, Kearney, and Ernst (1997) showed that interviewers using more anonymous techniques for forming household rosters improve the coverage of selected subgroups (e.g., young black males), suggesting that variability among interviewers in rostering procedures could produce variance in coverage rates. Martin (1999) suggested that variability among interviewers in the ability to precisely clarify who should be counted as a household member could also lead to varying coverage errors. SAMHSA (2008) found that interviewers with more face-to-face experience successfully screened higher proportions of eligible dwelling units. Consistent with Eckman and O'Muircheartaigh (2011), Tourangeau, Kreuter, and Eckman (2012) suggested that some interviewers may report that difficult households are not eligible and found that interviewers with higher screener completion rates tended to find fewer eligible households.

Variance among interviewers in coverage rates could therefore be introduced by variability in the importance placed on high response rates versus high eligibility rates, rostering procedures used, and procedures used to build address lists. Future research should operationalize these three factors (e.g., percentage of household rosters using anonymous/abbreviated listings) and examine their associations with coverage rates *and* coverage errors.

3. EXPLAINING INTERVIEWER EFFECTS ON UNIT NONRESPONSE ERROR

3.1 Experience

Interviewers need to make contact with (and gain cooperation from) sampled cases. Purdon, Campanelli, and Sturgis (1999) suggest that interviewers with more experience working on a *given survey* employ more effective strategies for making contact and that more *overall* experience is associated with higher contact rates. Lievesley (1986) reported that interviewers who hold other jobs, are willing to travel, and are willing to work weekends tend to have higher contact rates, but Beerten (1999) contradicted the "other job" effect.

Table A.1 in the online appendix summarizes several studies that have examined the relationship of experience with gaining cooperation *given contact*. Immediately apparent is the wide array of research designs and measures of experience used. Collectively, these studies suggest that more experience tends to increase the probability of cooperation (table 2), but there are certainly exceptions; these effects may also be moderated by respondent or area features (Mierzwa, McCracken, Vandermaas-Peeler, and Tronnier 2002).

The conceptual mechanisms underlying these positive relationships are still not well-understood (Couper and Groves 1992; Jäckle, Lynn, Sinibaldi, and Tipping 2013). Groves and Couper (1998) suggest that more experienced interviewers are better at tailoring their approaches to the type of resistance faced (Lemay and Durand 2002) and maintaining interaction with potential respondents. Per Jäckle et al. (2013), "One fifth of the effect of experience on cooperation is explained by differences in [traits/skills/attitudes]," raising the possibility that interviewers with less favorable characteristics for recruiting respondents are more likely to leave the workforce. Similar attempts to understand other mediators of these positive relationships are still needed, especially given repeated evidence of null and curvilinear relationships. While these findings may "[reflect] the common strategy that more successful and experienced interviewers are assigned to the more difficult cases" (Blom, de Leeuw, and Hox 2011), experienced interviewers may also get increased workloads, leading to lower response rates (Singer, Frankel, and Glassman 1983; Loosveldt, Carton, and Pickery 1997; Japec 2008) and limited ability to attempt contact during more productive times (Bottman and Thornberry 1992).

Table 2 Distributions of Results from Related Studies Examining the Effects of Interviewer Characteristics on Response Rates (see Table A.1 in the Online Appendix for Study Details)

Interviewer characteristic	Number of studies identified	Distribution of primary results
Experience	25	**Significant or marginally significant positive relationship: 15 studies** Null findings: 6 studies Curvilinear relationship (suggesting that positive effects of experience are limited to initial experience gained): 3 studies Negative relationship (when adjusting for pay grade): 1 study
Age	14	**Null findings: 7 studies** Weak positive relationship: 4 studies Curvilinear or nonmonotonic relationships: 3 studies
Gender	15	**Null findings: 11 studies** Marginal or significant positive effect of being female: 3 studies Gender effects moderated by respondent features: 1 study
Optimism/positive attitudes/ self-confidence	12	**Positive relationships: 12 studies**

NOTE.—**Boldfaced** results indicate the most common (or modal) finding.

3.2 Sociodemographic Characteristics

Purdon, Campanelli, and Sturgis (1999) also found that men were more comfortable than women with making a large number of evening calls, which tends to be an effective contact strategy (Blohm, Hox, and Koch 2007), but women and older interviewers have been reported to achieve higher contact rates (Blohm et al. 2007). Studies analyzing the effects of gender and age on cooperation (table A.1) have generally found weak or nonsignificant relationships (table 2). The evidence presented in tables A.1 and table 2 is consistent with the overall conclusions of Morton-Williams (1993) and Groves and Couper (1998) that the sociodemographic characteristics of interviewers do not tend to have substantial main effects on cooperation rates.

Some studies suggest that "liking" (or sociodemographic similarities between the interviewer and the respondent) increases cooperation rates. Durrant, Groves, Staetsky, and Steele (2010) found that matching based on gender and education tended to produce higher cooperation rates, and Brehm (1993), Webster (1996), Moorman, Newman, Millikan, Tse, and Sandler (1999), and Lord, Friday, and Brennan (2005) also reported that individuals contacted by interviewers with similar demographic features were more likely to cooperate. However, "liking" may not be feasible in larger studies, does not always improve cooperation rates (Wang, Kott, and Moore 2013), and (as we discuss later) may increase measurement error.

3.3 Attitudes and Personalities

Many survey organizations ask interviewers to complete questionnaires about their *attitudes* regarding surveys (de Leeuw and Hox 2009; Blom and Korbmacher 2013), and several studies have examined whether these attitudes predict contact and cooperation (table A.1). Blom et al. (2011) found that interviewers valuing a positive and professional image had higher contact rates (see also Stefan, Jacob, and Gueguen 2015) and that interviewers having more positive attitudes toward persuasion, responding to reluctance in a more optimistic fashion, or having higher self-confidence in general have consistently produced higher cooperation rates (table 2). Maynard and Schaeffer (2002a) arrive at similar conclusions when analyzing transcripts of telephone survey requests. These relationships become even stronger for interviewers with less experience (Jäckle et al. 2013) and in more complex surveys (Durrant et al. 2010). Interviewers with more positive attitudes will tend to have higher cooperation rates.

More recently, studies have examined the effects of different interviewer *personalities* on unit nonresponse. Jäckle, Lynn, Sinibaldi, and Tipping (2013) found that assertive, extroverted, "less-open" (i.e., less accepting of new ideas) interviewers had higher response rates. Vassallo, Durrant, Smith, and Goldstein (2015) reported that no tendencies to worry, less complacency about the ability to deliver a survey's message, greater awareness of the way that interviewers portray themselves, and making a conscious effort to communicate produced higher response rates. Yu (2014) also reported evidence of extroverted interviewers having higher response rates and more agreeable (or "respondent-centered") interviewers having lower cooperation rates, similar to Jäckle, Lynn, Sinibaldi, and Tipping (2013) and Snijkers, Hox, and de Leeuw (1999). Despite these common findings, some of these relationships are unexpected (e.g., agreeableness and being more accepting of new ideas having negative effects), motivating additional research in this area.

3.4 Skills and Behaviors

Several studies have examined the relationships of interviewer behaviors and skills with unit nonresponse. Campanelli, Sturgis, and Purdon (1997) reported that interviewer *persistence* in making follow-up calls decreased noncontact rates.

Purdon, Campanelli, and Sturgis (1999) found that interviewers who make calls during weekday evenings generally have better chances of making contact, and Weeks, Kulka, and Pierson (1987), Groves and Couper (1998), and Blom (2012) reported that interviewer variance in calling times can predict interviewer variance in contact rates. These findings suggest that the effects of experience on contact rates reported by Purdon, Campanelli, and Sturgis (1999) may be *mediated* by persistence (i.e., experience increases persistence, and persistence increases contact rates). Multiple studies have also demonstrated that interviewers with higher contact rates also have higher cooperation rates (O'Muircheartaigh and Campanelli 1999; Pickery and Loosveldt 2002; Blom et al. 2011; Durrant and D'Arrigo 2014). These findings suggest that higher-performing interviewers have the skills necessary to master both tasks (e.g., persistence for making contact and tailoring interactions for securing cooperation; Morton-Williams 1993; McCarty, House, Harman, and Richards 2006; see also Link 2006).

Many studies have utilized audio recordings of the survey introduction, either on the doorstep or the telephone, to better understand variability in interviewer behaviors during this important interaction. Studies of the vocal characteristics and speech patterns of interviewers have consistently found that moderate speech characteristics (e.g., not speaking too fast or too slow) during the introduction increase cooperation rates (Oksenberg, Coleman, and Cannell 1986; Oksenberg and Cannell 1988; Groves, O'Hare, Gould-Smith, Benki, and Maher 2008; Conrad, Broome, Benki, Kreuter, Groves, et al. 2013). Morton-Williams (1993) found that interviewers who used concise introductions, maintained interaction, used social skills to tailor responses to reluctance, and used friendlier introductions produced higher response rates. These findings were generally echoed in studies by Campanelli et al. (1997), Sturgis and Campanelli (1998), de Leeuw (1999), Snijkers et al. (1999), and Houtkoop-Steenstra and van den Bergh (2002), which collectively suggest that more experience on the doorstep may engender the skills needed (tailoring ability, maintaining interaction as defined by Groves, Cialdini, and Couper 1992, and persuasion) to effectively convince respondents to cooperate (i.e., these skills may *mediate* the effects of experience on cooperation rates).

Empirical evidence of these three skills producing higher cooperation rates was later found in studies using experimental methods (Groves and McGonagle 2001), observational methods (Dijkstra and Smit 2002; Couper and Groves 2002; Maynard and Schaeffer 2002b; Durrant et al. 2010; Broome 2015), and conversation analytic methods (Maynard, Freese, and Schaeffer 2010). Schaeffer, Garbarski, Freese, and Maynard (2013) suggested that interviewers should always introduce themselves, try to reduce the burden of the interview request, and use "graceful exits" tailored to any negative statements (e.g., "I'm sorry that you aren't interested right now; perhaps we can try again later" rather than "Why are you not interested?"), which may improve refusal conversion efforts. Interviewers with more of these skills may have higher rates of cooperation and (in turn) higher pay (de Heer 1999; Durrant et al. 2010), but they may also be selected and retained by survey organizations at higher rates.

3.5 Obtaining Additional Consent

Some surveys task interviewers with obtaining respondent consent for additional data collection during the interview. Studies of interviewer effects on this process have generally found high interviewer variability in consent rates (e.g., Beste 2011). Sala, Burton, and Knies (2012) found that more experience with the current survey resulted in higher consent rates, but Sakshaug, Couper, and Ofstedal (2010) and Sakshaug, Couper, Ofstedal, and Weir (2012) were unable to identify strong explanatory factors. Korbmacher and Schröder (2013) and Sakshaug, Tutz, and Kreuter (2013) found *negative* effects of experience on consent rates, and interviewer attitudes may also matter: "Interviewers who themselves would be willing to consent to data linkage requests were more likely to obtain linkage consent from respondents" (Sakshaug et al. 2013). Collectively, this small literature suggests that interviewer experience and attitudes might explain interviewer effects on consent, but the results of these studies have been inconsistent. Further, no studies have examined interviewer effects on consent *bias*.

3.6 A Critical Knowledge Gap: Interviewer Effects on Nonresponse Bias

No studies to date have attempted to identify factors explaining interviewer effects on nonresponse *bias*, which is generally a difficult endeavor (West and Olson 2010; West, Kreuter, and Jaenichen 2013; Loosveldt and Beullens 2014). Blohm et al. (2007) emphasize that variability in response rates among interviewers will not necessarily imply variability in nonresponse bias (see also Groves 2006). For example, an interviewer who obtains a relatively high response rate by only recruiting older sampled individuals may introduce more nonresponse bias than an interviewer with a lower response rate who recruits a smaller but more diverse set of sampled individuals. In addition, this variability in the features of recruited individuals could introduce what seems like interviewer variance in the survey measures collected (Stock and Hochstim 1951; Moser and Stuart 1953; Tucker 1983; Stokes and Yeh 1988; West and Olson 2010; West et al. 2013).

4. EXPLAINING INTERVIEWER EFFECTS ON MEASUREMENT ERROR

After successful recruitment of sampled units, interviewers are tasked with conducting interviews and collecting survey data, and there has been a rich history of research examining sources of interviewer variability in survey measures. We first consider the types of survey questions that are prone to interviewer effects and then we consider the effects of interviewer characteristics on survey measurement.

4.1 Survey Questions Prone to Interviewer Effects

Interviewer effects are estimate-specific and therefore may depend on the features of survey questions. Schaeffer, Dykema, and Maynard (2010) provide a comprehensive review of this literature, which generally suggests that, outside of a few exceptions (Kish 1962; Fellegi 1964; Groves and Magilavy 1986; Mangione, Fowler, and Louis 1992), attitudinal, sensitive, ambiguous, complex, and open-ended questions are more likely to produce variable interviewer effects (table 3). These types of questions introduce more opportunities for interviewers to intervene or assist respondents (Groves and Magilavy 1986; Billiet and Loosveldt 1988; Schnell and Kreuter 2005). Poorly designed instruments can also increase the burden that an interviewer perceives for a given survey, which will likely have negative effects on both response rates and measurement quality (Houtkoop-Steenstra 2000; Japec 2005, 2008). The specific interviewer characteristics, behaviors, and skills that explain the variability in interviewer effects for these specific types of questions (e.g., variation in experience and prior training [Feldman, Hyman, and Hart 1951; Billiet and Loosveldt 1988; Olson and Smyth 2015], expression of nonverbal signals [Holbrook, Green, and Krosnick 2003], failure to follow directions [Gray 1956; O'Muircheartaigh 1976], interviewer beliefs [e.g., Himelein 2015], and variation in interviewing styles [van der Zouwen, Dijkstra, and Smit 1991]) have been the subject of a large body of research that we now consider.

4.2 Race/Ethnicity

Dating back to the work of Rice (1929), who found that individuals interviewed by supporters of prohibition attributed their alcohol dependence to the liquor itself rather than other "industrial" factors (e.g., plant shutdowns), a large body of literature has examined the relationships of sociodemographic characteristics of interviewers with survey responses (Sudman and Bradburn 1974; Schaeffer et al. 2010). Some of these studies have used experimental approaches to examining these relationships, following the original studies performed by Hyman, Cobb, Feldman, Hart, and Stember (1954) examining race-of-interviewer (ROI) effects.

Many studies have examined the relationships of interviewer race or ethnicity with responses to various survey items (Davis, Couper, Janz, Caldwell, and Resnicow 2010; see table A.2 in the online appendix). These studies have employed a rich variety of designs, survey measures, and analytic approaches, but

Table 3 Features of Survey Questions Repeatedly Demonstrated to Introduce Interviewer Effects

Type of survey question	Studies demonstrating interviewer effects
Attitudinal/opinion/ ambiguous/subjective	Stock and Hochstim (1951); Franzen and Williams (1956); Gray (1956); Hanson and Marks (1958); O'Muircheartaigh (1976); Collins and Butcher (1982); Davis and Scott (1995); Schnell and Kreuter (2005); Davis, Couper, Janz, Caldwell, Resnicow, et al. (2010); Brunton-Smith, Sturgis, and Williams (2012); Himelein (2015)
Open-ended	Feldman, Hyman, and Hart (1951); Gray (1956); Sudman and Bradburn (1974); O'Muircheartaigh (1976); Collins (1980); Groves and Magilavy (1986); Mangione , Fowler, and Louis (1992); Schnell and Kreuter (2005)
Emotional/sensitive	Ellis (1947); Stock and Hochstim (1951); Bailey, Moore, and Bailar (1978); Bignami-Van Assche, Reniers, and Weinreb (2003); Schnell and Kreuter (2005)
Difficult/complex skip patterns	Gray (1956); Collins (1980); Collins and Butcher (1982); Brüderl, Huyer-May, and Schmiedeberg (2013); Josten and Trappmann (2016)
Race or gender-related	See table A.2 in the Online Appendix

NOTE.—The studies in table 3 generally did not focus on explanations for the effects; we simply provide a sense of the number of studies repeatedly demonstrating that these questions introduce interviewer effects and consider this to be motivation for studies of what *explains* the interviewer effects for these *specific* types of questions.

the various effects reported in these studies consistently tell the same qualitative story (table 4): ROI effects are strongly moderated by respondent race and the racial sensitivity of the question being asked, and respondents will strive to provide (possibly inaccurate) answers that will not offend an interviewer of the race in question, even in *self-administered* modes (Liu and Wang 2016). These

Table 4 Distributions of Results from Related Studies Examining the Effects of Interviewer Characteristics on Response Quality (see Table A.2 in the Online Appendix for Study Details)

Interviewer characteristic	Number of studies identified	Distribution of primary results
Race/ethnicity	30	**Different response distributions for racially sensitive questions (less offensive responses to interviewers likely to be offended): 17 studies (out of 18 studies with racially sensitive questions)**
		More socially desirable responses for nonsensitive questions depending on race/ethnicity: 9 studies (out of 16 studies with nonsensitive questions)
		Differences in responses moderated by respondent race: 10 studies (out of 10 studies examining moderation)
		Generally null findings: 5 studies
Age	24	**Null findings: 9 studies**
		Positive relationship with response quality: 5 studies
		Negative relationship with response quality: 3 studies
		Relationship moderated by respondent/interviewer features: 3 studies
		Significant but unclear relationships: 2 studies
		Curvilinear relationship: 2 studies
		Age moderates the relationships of other features: 1 study
Sociodemographic matching with respondent characteristics	10	**Positive effect on response quality: 4 studies**
		Negative effect on response quality: 3 studies
		Null findings: 2 studies
		Matching increases interviewer variance: 1 study
Overall experience	12	Positive relationship with response quality: 4 studies
		Negative relationship with response quality: 4 studies
		Null findings: 4 studies
Survey-specific experience	5	Negative relationship with response quality: 2 studies
		Null findings: 2 studies
		Positive relationship with response quality: 1 study
Current survey experience	6	Positive relationship with response quality: 2 studies
		Negative relationship with response quality: 2 studies
		Curvilinear relationship with response quality: 1 study
		Null findings: 1 study
Gender	23	**Null findings: 10 studies**
		Females collect higher-quality responses: 9 studies
		Males collect higher-quality responses: 4 studies
		Gender effects moderated by respondent gender: 3

NOTE.—Distributions of results in table 4 do not necessarily sum to the total number of studies because multiple types of findings (or lack thereof) may have been reported in a single study. **Boldfaced** results indicate the most common (or modal) finding.

effects are all consistent with the theory of Zaller and Feldman (1992) that people will respond to survey questions "on the basis of ideas that happen to be, for whatever reason, immediately salient to them" (p. 603). While much of this literature suggests that ROI effects arise due to this type of "interpersonal deference" (e.g., Athey, Coleman, Reitman, and Tang 1960; Schuman and Converse 1971; Campbell 1981; Reese, Danielson, Shoemaker, Chang, and Hsu 1986; Webster 1996; Davis 1997a), some studies have presented evidence of negative responses *not* being censored by the presence of an interviewer having a different race, attributing these findings to the interviewer's race activating negative stereotypes (Devine 1989; Davis and Silver 2003; Krysan and Couper 2003; Krysan and Couper 2006).

Many of these studies have called for matching of respondents and interviewers by race (e.g., Williams 1964; Worcester and Kaur-Ballagan 2002; Cobb, Boland-Perez, and LeBaron 2008) or sociodemographic features more generally in order to improve measurement quality, under the theory that respondents may edit their responses to be consistent with perceived normative expectations of an interviewer having *different* features (Johnson, Fendrich, Shaligram, Garcy, and Gillespie 2000). The studies summarized in tables A.2 and table 4 suggest that while this strategy can improve measurement quality, it can also backfire (e.g., Anderson, Silver, and Abramson 1988a; see also Rhodes 1994 and Davis, Caldwell, Couper, Janz, Alexander, et al. 2013). Placing more weight on the studies using direct measures of similarity and response quality in table A.2, we find slightly more evidence in support of the matching approach.

More work is needed to further understand potential mediators of these ROI effects, such as workload (Cody, Davis, and Wilson 2010). Specifically, respondent self-administration of questions related to the racial stereotypes invoked by an interviewer and efforts to avoid offending the interviewer with responses to racially sensitive questions would help to formally test the possible mediating effects of "deference" and "stereotype activation" suggested in the literature. The identification of mediators of these effects could help with the development of different strategies aside from the matching approach for reducing ROI effects (e.g., reminding respondents that their answers will not be considered offensive in any way).

4.3 Gender

Several studies have examined gender-of-interviewer (GOI) effects on response quality (Davis et al. 2010) (table A.2). The summaries in table A.2 show that, as with ROI effects, the type of gender-related question being asked generally moderates GOI effects, with female interviewers having a tendency to collect higher-quality responses than males, and several studies producing

Table 5 Distributions of Results from Related Studies Examining the Effects of Alternative Interviewer Behaviors on Response Quality (see Table A.3 in the Online Appendix for Details)

Interviewer behavior	Number of studies identified	Distribution of primary results
Providing feedback/ probing	6	**Positive relationship with response quality: 3 studies** Moderation of the effects of other interviewer features on quality: 2 studies Negative relationship with response quality: 2 studies
Using conversational/ flexible/personal interviewing styles (versus standardized/ formal/professional interviewing styles)	9	**Positive relationship with response quality (for complex items): 8 studies** Negative relationship with response quality: 1 study
Efforts to build rapport	8	**Negative relationship with response quality: 5 studies** Null findings: 3 studies

Note.—Distributions of results in table 5 do not necessarily sum to the total number of studies, because multiple types of findings (or lack thereof) may have been reported in a single study. **Boldfaced** results indicate the most common (or modal) finding.

null findings (table 4). When considering the alternative hypotheses that GOI effects could occur due to either deference to the interviewer or priming of stereotypes (Tourangeau, Couper, and Steiger 2003), we found that while some of the papers suggested deference (e.g., Kane and Macaulay 1993; Liu and Wang 2016) or even "antideference" (e.g., Landis, Sullivan, and Sheley 1973) explanations, more of the papers finding GOI effects spoke to the activation of stereotypes. Females may be perceived as less threatening (Axinn 1989) and may elicit different considerations than males for gender-related questions (Huddy, Billig, Bracciodieta, Hoeffler, Moynihan et al. 1997), possibly resulting in more socially desirable responses (Liu and Stainback 2013). Males have collected more frequent reports of drug use and chronic conditions, possibly because respondents perceive them as more likely to use drugs (Johnson and Parsons 1994) or more "doctor-like" (Edwards and Berk 1993). Some work has suggested that the intonation of an interviewer's voice may be a reason for GOI effects (Barath and Cannell 1976; see also Fahrney, Uhrig, and Kuo 2010 and Dykema, Diloreto, Price, White, and Schaeffer 2012), but these findings have been disputed (Blair 1977). As with the ROI effect literature, these findings motivate future research on strategies for mitigating the effects of stereotype activation in surveys asking gender-related questions.

4.4 Age

Considering 24 studies that have examined the relationships of interviewer age with various indicators of response quality (table A.2), the modal finding was a null relationship (table 4). However, both positive and negative relationships have also been reported, and alternative hypotheses for these relationships have been presented (but not tested empirically). Older interviewers have struggled to collect complete information, possibly due to a negative relationship between age and reading comprehension or ability to follow instructions (Hanson and Marks 1958; see also Brüderl, Huyer-May, and Schmiedeberg 2013 and Josten and Trappmann 2016), which could lead to longer interviews (Olson and Bilgen 2011). Other studies have found opposite relationships of age with item-missing data (Berk and Bernstein 1988), however, and studies reporting positive age relationships have suggested increased experience (Sudman and Bradburn 1974) and more comfort with the interviewing situation (Cleary, Mechanic, and Weiss 1981) as mediating factors. Older interviewers may also be viewed as authority figures, leading to edited responses that are more socially desirable or consistent with what respondents believe those in authority would expect to hear (Ehrlich and Riesman 1961; Johnson and Parsons 1994). These relationships may also be limited to certain subgroups thinking that particular types of responses are indeed socially desirable (Reinecke and Schmidt 1993). Interviewer age has also been shown to moderate the effects of other interviewer features (Freeman and Butler 1976; Moum 1998), meaning that analyses of response quality should consider interactions of interviewer age with other characteristics.

4.5 Experience

The effects of different types of interviewer experience (overall, survey-specific, or current-survey) on response quality have been found to vary widely (table A.2), with few clear patterns evident in the literature (see tables A.2 and table 4). Similar to other studies of interviewer characteristics, we found that the studies testing experience effects varied widely in terms of the multivariable models used to examine the effects, with different covariates employed at both the interviewer and respondent levels. If important mediators of the relationship of a characteristic like experience with survey responses (e.g., length of administration; see section 4.8 below) are accounted for in some models but not others, there may be null findings in one study (mediators included) but significant findings in another (mediators excluded). Better understanding of potential mediators of these relationships and standardization of the modeling approaches used is clearly still needed in this area.

4.6 Physical Appearance

Eisinga, te Grotenhuis, Larsen, Pelzer, and van Strien (2011) found that interviewers with higher body mass index (BMI) values tended to collect responses suggesting more restraint in eating, although opposite results were found in a study varying the physical appearance of *virtual* interviewers (Murphy, Dean, Cook, and Keating 2010). Bateman and Mawby (2004) found that more formally dressed interviewers collected stronger indicators of willingness to pay for an environmental good, Bailar, Bailey, and Stevens (1977) found item nonresponse on income questions to be higher among interviewers with higher incomes and interviewers opposed to asking income questions, and Riessman (1979) found that female respondents were less likely to report psychiatric symptoms to high-status (physician) interviewers. Collectively, these studies suggest that interviewers appearing to have higher socio-economic status may ultimately collect data of lower quality, especially if the questions are related to the physical appearance of the interviewer.

4.7 Behaviors and Skills

Variability in interviewer behaviors during administration may also lead to variance in response distributions (Schaeffer and Dykema 2011). Hanson and Marks (1958) found that certain interviewers may omit (or assume responses to) questions that they "resist" and that some interviewers may probe more than others. Cannell, Oksenberg, and Converse (1977) demonstrated that interviewers can affect the complex cognitive process that survey respondents follow when answering survey questions (Tourangeau, Rips, and Raskinski 2000) via the behaviors that they use (Cannell, Miller, and Oksenberg 1981). Several studies have suggested that respondents may react to nonverbal signals sent by face-to-face interviewers, especially for sensitive or attitudinal items, and varying interviewer behaviors introduced by alternative modes of data collection (e.g., different CAI interfaces) may ultimately affect respondent behaviors (Fuchs 2002; Holbrook et al. 2003). Despite these possibilities, studies using validation data have suggested that *respondent* behaviors have more of a consistent association with response accuracy than *interviewer* behaviors in a standardized interview setting (Dykema, Lepkowski, and Blixt 1997). The effects of interviewer behaviors like excessive probing and providing feedback on the adequacy of responses have generally been mixed (see tables A.3 and table 5), but these behaviors are likely reactions to poor respondent behavior (Belli and Lepkowski 1996) and may simply serve to increase the risk of interviewer variance due to other factors (Mangione et al. 1992).

Variability among interviewers in the techniques that they use to administer questions and address respondent confusion (or requests for clarification; see Sander, Conrad, Mullin, and Herrmann 1992 and Schaeffer and Maynard 2002) may also lead to variance in response accuracy among interviewers. "Standardized Interviewing" (Fowler and Mangione 1989) is often recommended for minimizing interviewer effects, given the complex interactions that can arise between respondents and interviewers (Ongena and Dijkstra 2007). However, interviewers occasionally deviate from strictly standardized interviewing, either by mistake (e.g., leaving out words) or to make interactions with respondents more cohesive (Peneff 1988; Houtkoop-Steenstra 1995; Haan, Ongena, and Huiskes 2013; Ackermann-Piek and Massing 2014; Bell, Fahmy, and Gordon 2016). Both experimental and observational studies have repeatedly suggested that more personal, flexible interviewing techniques designed to ensure respondent comprehension may reduce response bias for complex factual questions (see tables A.3 and table 5; Belli, Lee, Stafford, and Chou 2004 present exceptions, which again may be due to poor respondent behavior).

Whether more flexible techniques introduce more interviewer variance for particular types of items (e.g., Sayles, Belli, and Serrano 2010), possibly offsetting increases in estimate quality due to bias reduction, is an important open question. Formulation of effective follow-up probes is the result of a taxing cognitive process (Japec 2008). Houtkoop-Steenstra (1997) and van der Zouwen, Dijkstra, and Smit (1991) suggest that more personal interviewing styles may result in differential use of reformulated neutral probes that ultimately lead respondents. Further, more frequent use of verification probes (Moore and Maynard 2002) or suggestive probes (Smit, Dijkstra, and van der Zouwen 1997) by interviewers upon hearing inadequate answers may lead to variance in measurement. More research examining these bias-variance tradeoffs depending on the technique used is still needed.

Interviewers who deviate excessively from general interviewing scripts in an effort to build rapport, possibly defined as the development of a harmonious relationship with the respondent (Goudy and Potter 1975), have repeatedly been shown to either produce lower response accuracy or have no effect on accuracy (Gabarski, Schaeffer, and Dykema 2016; tables A.3 and table 5). There are many ways to define and operationalize "rapport building" (Belli, Lepkowski, and Kabeto 1999; see also Lavin and Maynard 2002), which could be viewed as a form of tailoring, and the combination of this conceptual weakness with early evidence of interviewer variance in the ability to even *establish* rapport (Feldman et al. 1951) makes this a difficult topic to study. Interviewers with greater skill at addressing respondent concerns during *recruitment* have been shown to have smaller effects on certain response distributions (Brunton-Smith, Sturgis, and Williams 2012), but this literature consistently suggests that excessive variability in the effort to establish rapport during *measurement* will likely lead to more variability in response distributions.

Interviewers may also behave differently during the self-administered portions of face-to-face interviews. While one would expect interviewer effects to be negligible in self-administered modes relative to interviewer-administered modes (Tourangeau, Rasinski, Jobe, Smith, and Pratt 1997), several studies have reported surprising evidence of non-zero interviewer effects on responses to *self-administered* survey questions (Summers and Hammonds 1966; O'Muircheartaigh and Wiggins 1981; O'Muircheartaigh and Campanelli 1998; Hughes, Chromy, Giacoletti, and Odom 2002; RTI 2012; West and Peytcheva 2014; Liu and Wang 2016). Interviewers may vary in their ability to establish private settings for the interview (Aquilino 1993; Herzog 2005; Mneimneh 2012), provide different types of help during the self-administration (Couper and Rowe 1996; Mneimneh, Tourangeau, Heeringa, and Elliott 2014), and introduce "bystander" effects due to their physical location during the self-administration (Aquilino 1993; Weisband and Kiesler 1996; Aquilino, Wright, and Supple 2000; Santtila, Korkman, and Sandnabba 2004; Kroh 2005; West and Peytcheva 2014). Future studies linking these behaviors to variability in response distributions will benefit future interviewer training.

4.8 Interviewer Effects on Paradata

Interviewers with more experience (survey-specific or overall) have been found to record observations of household features with *lower* accuracy (West and Kreuter 2013, 2015), but not consistently (Sinibaldi, Durrant, and Kreuter 2013). Interviewer race has also been found to influence interviewer perceptions of skin color (Hill 2002). Interviewers may use different strategies when recording these types of observations (West and Kreuter 2011; West, Kreuter, and Trappmann 2014; West, Li, Ma, and Wang submitted), reasons for refusals (Menold and Zuell 2016), and *post-survey* observations describing respondent behavior, which have repeatedly been shown to vary substantially across interviewers (Feldman et al. 1951; Cleary et al. 1981; O'Muircheartaigh and Campanelli 1998; Kaminska, McCutcheon, and Billiet 2010). These findings motivate the development of standardized training and future research on the processes that interviewers follow when recording these observations.

Some studies have also assessed interviewer effects on interview length (Olson and Peytchev 2007; Olson and Bilgen 2011; Couper and Kreuter 2013; Olson and Smyth 2015), finding that male interviewers, younger interviewers, and interviewers with more overall experience tend to produce *shorter* interviews (especially early in a field period), which could in turn have higher rates of measurement error (Loosveldt and Beullens 2013). Interviewers with more experience working on a given survey that stresses quantity (completed interviews) over quality (careful responses from respondents) may take shortcuts or give subtle hints to respondents that affirmative responses to specific questions may lengthen the survey, limiting their ability to finish more interviews (e.g., Brüderl et al. 2013).

5. INTERVIEWER EFFECTS ON PROCESSING

Several studies have presented evidence of variability among interviewers in their ability to successfully enter and code open-ended respondent answers during measurement (Collins and Courtenay 1985; Crul, Van Valey, Johnson, Ayuso, Bartz, et al. 1991; Houtkoop-Steenstra 1996; Hak 2002; Krosnick, Lupia, DeBell, and Donakowski 2008; Sayles et al. 2010). Similar coding operations performed by professional coders *after* fieldwork tend to produce negligible between-coder variance (Kalton and Stowell 1979; Campanelli, Thomson, Moon, and Staples 1997), although larger workloads might be detrimental (Sturgis 2004). Very little research has sought to identify clear reasons for the variability introduced by interviewers during "live" field coding. For example, might interviewers with more experience be less likely to make these types of errors?

Smyth and Olson (2015) found that some interviewers mistype, mishear, or simply enter wrong information at non-negligible rates when coding open-ended responses into nominal variables, especially when probing and providing feedback. Variable propensity to make these types of coding errors could ultimately affect skip patterns depending on prior answers, potentially resulting in variable measurement errors. Future research in this area needs to build on the work of Smyth and Olson (2015) and Krosnick et al. (2008), analyzing audio recordings of open-ended responses and interviewer characteristics to identify the factors that lead to variable processing errors among interviewers.

6. CONCLUSION: AN ORGANIZING MODEL FOR FUTURE RESEARCH

This research synthesis generally finds that *background characteristics* of the interviewers, including sociodemographic features and experience, affect various indicators of survey errors (e.g., interviewers with more experience generally tend to produce higher response rates). The synthesis also suggests that other variables may *mediate* these relationships—including interviewer attitudes (e.g., optimism), personalities, behaviors (e.g., probing, speed), and skills (e.g., tailoring, persistence)—and also activation of stereotypes or perceived norms in the respondents. These mediating variables are potentially influenced by the background characteristics and, in turn, affect the various indicators of errors. The synthesis further finds consistent evidence of the effects of these background characteristics and mediators being *moderated* by the features of a survey question, respondent characteristics (e.g., Riessman 1979), or other interviewer characteristics (e.g., Moum 1998). For example, interviewers with different race/ethnicity backgrounds or genders tend to collect different response distributions for sensitive questions about race or gender, but these effects may not hold for less-sensitive factual questions. Furthermore, "liking" studies suggest that matching sociodemographic features of interviewers and respondents is generally beneficial for increasing unit response rates, but not necessarily for high-quality measurement.

Although the literature has provided consistent support for some of the relationships mentioned above, many studies have produced null findings related to the relationships of particular interviewer characteristics with selected indicators of survey error, failing to fully explain interviewer variance in those indicators. Explaining interviewer effects is clearly a challenging task, requiring a structured approach that builds on existing knowledge. We therefore conclude by describing an organizing model for future studies attempting to explain interviewer effects on multiple error sources that is motivated by this research synthesis (figure 1). The proposed model also draws on previous models suggested for explaining interviewer effects (Dijkstra and van der Zouwen 1987, p. 206; van der Zouwen et al. 1991; Reinecke and Schmidt 1993; Japec 2008).

Our organizing model in figure 1 focuses on potential mediators of interviewer effects on survey errors. This approach is motivated by the outcomes of several of our syntheses, which revealed both null findings and significant direct effects when considering the relationship of a particular interviewer characteristic with a specific indicator of survey error (e.g., relationships of interviewer age with response quality). The inclusion of important mediators in some models but not others may lead to different findings regarding a particular relationship. Our organizing model can be applied to understand what is already known about interviewer effects on various error sources and to discover gaps in knowledge that motivate future studies. We now consider some of these potential applications.

6.1 Applications of the Organizing Model

Example 1. The synthesis reports various studies examining the relationship between interviewer experience and nonresponse. Most studies find that more experience engenders higher response rates. However, many studies also report null findings or even negative relationships. Furthermore, multiple studies have suggested that more experience leads to more skill at tailoring responses to expressed concerns (possibly measured as the number of *unique* responses to concerns used, from audio recordings) and maintaining interaction with units, and interviewers with more skills in these areas (who may also be retained at higher rates by survey organizations) have repeatedly been shown to have higher responses rates. The synthesis therefore suggests that these skills

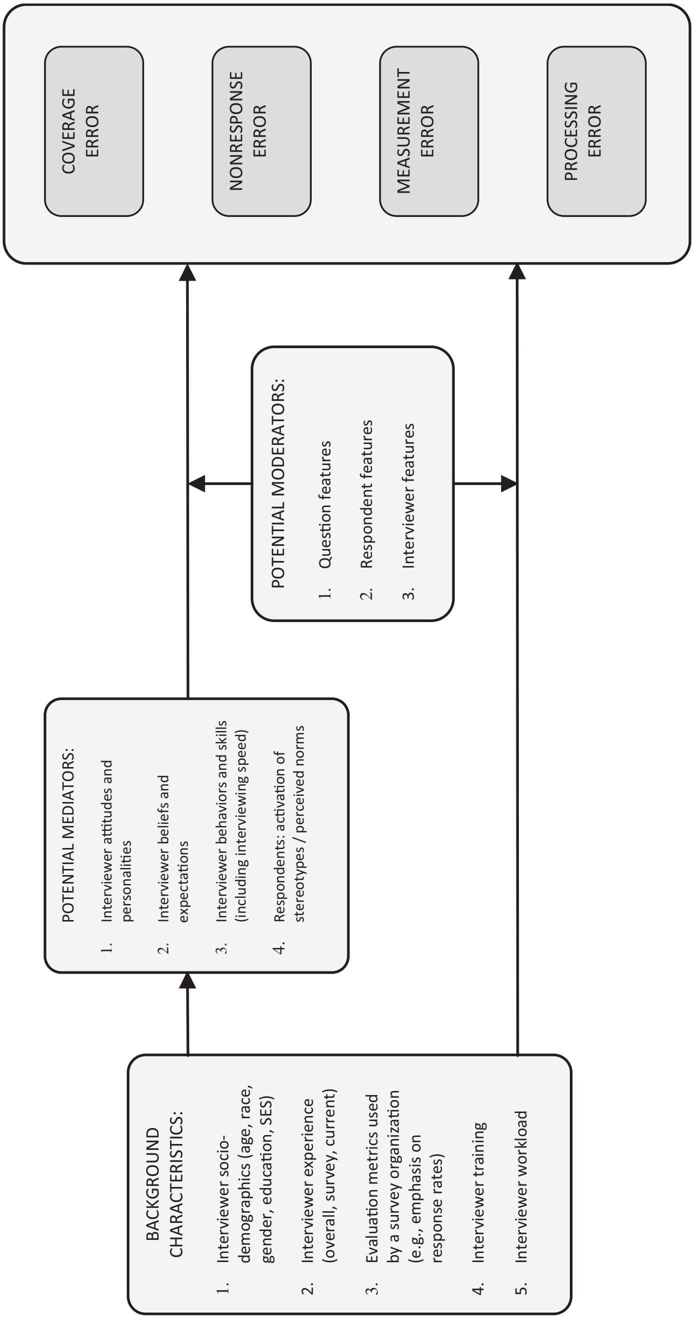

Figure 1. An Organizing Model for Research Attempting to Explain Interviewer Effects.

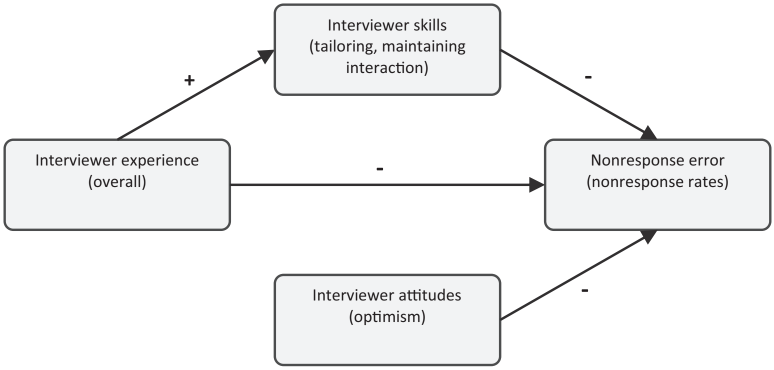

Figure 2. "Zooming in" on a Portion of the Organizing Model in Figure 1 to Understand What is Known about the Relationship of Interviewer Experience with Nonresponse Error (+ Indicates Positive Relationship, – Indicates Negative Relationship).

may mediate the relationship of experience with response rates (figure 2). Differences across studies in accounting for these mediating factors may explain the differences in relationships between interviewer experience and unit nonresponse found in the literature.

Example 2. Open-ended questions are prone to interviewer effects, and both Feldman, Hyman, and Hart (1951) and Olson and Smyth (2015) have suggested that more experienced interviewers may record richer responses to these questions. Following our model, one could design a study that tests persistence as a potential *mediator* of this experience effect (e.g., experienced interviewers may be more *persistent* when probing) by first operationalizing persistence (e.g., frequency of probing in audio recordings) and then assessing whether the experience effect remains when controlling for persistence in a model for open-ended response quality (e.g., number of words used). Considering interviewer effects on processing error, our model could also be applied to see if differential experience working on a *given survey* explains variance among interviewers in the accuracy of the codes that they enter describing open-ended responses, motivating additional practice with these "live" coding procedures for less-experienced interviewers.

Example 3. One can also use the organizing model to study possible *covariances* between different error sources introduced by interviewer effects. For example, the literature has suggested a possible *negative* covariance between nonresponse error and measurement error introduced by tailoring behaviors (Groves et al. 1992; Brunton-Smith et al. 2012): tailoring efforts during recruitment may increase response rates, but excessive efforts to establish rapport may *negatively* affect measurement quality (figure 3). Future studies could measure interviewer tailoring behaviors during both tasks (e.g., the number of off-topic statements made) and identify optimal behaviors for all interviewers

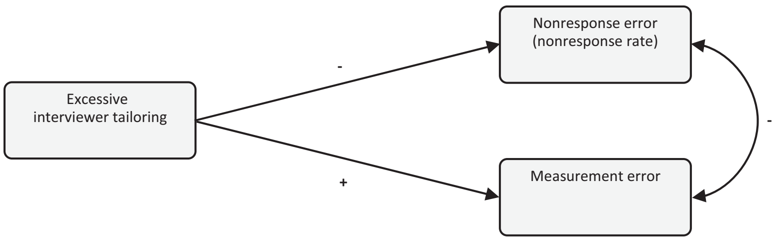

Figure 3. "Zooming in" on a Portion of the Organizing Model in Figure 1 to Understand What is Known about Interviewer Effects that May Introduce a Covariance Between Two Error Sources (e.g., Measurement Error and Nonresponse Error; + Indicates Positive Relationship, – Indicates Negative Relationship).

to engage in, eliminating this potential covariance introduced by interviewer variability in these behaviors.

In closing, we encourage survey researchers to apply the organizing model presented above when attempting to address important gaps in knowledge regarding explanations for interviewer effects. In particular, this synthesis has highlighted numerous opportunities for 1) identifying interviewer characteristics, skills, and behaviors that directly affect particular types of survey errors (e.g., coverage, processing) or 2) when the effects of interviewer characteristics on a particular error source are well-established (e.g., ROI effects on measurement error), identifying interviewer skills, behaviors, and attitudes that *mediate* the effects of the interviewer characteristics on that error source. Carefully designed studies identifying these sources of interviewer effects would improve future training sessions and ultimately benefit all data collections employing human interviewers.

Supplementary Materials

Supplementary materials are available online at http://www.oxfordjournals.org/our_journals/jssam/.

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